Satellite Conjunction Risk: a Statistical Analysis

What can an early risk estimate tell us about the final one?

The question

When two objects are expected to pass close to each other, operators receive updated conjunction data messages (CDMs). Each message gives a new estimate of the geometry, uncertainty and collision risk. The number can move substantially as tracking improves. This project asks how well an earlier message predicts the *last reported risk estimate* for the same encounter, and how much confidence we should put in that forecast.

That distinction matters: the dataset records calculated risk estimates, not collisions. A model that predicts the final CDM well has learned something about how estimates are revised. It has not learned the physical frequency of impacts.

The data and the target

The ESA Collision Avoidance Challenge archive contains 199,082 CDMs from 15,321 anonymised encounters. A typical encounter has 14 messages. The records include time to closest approach (TCA), current $\log_{10}$ risk, miss distance, relative speed, position-uncertainty fields, orbit information and a mission identifier.

For each encounter, I used the risk in its last CDM as the outcome. For regression, values below -15 were capped at -15 because the archive has a large artificial floor at -30. The separate warning task asks whether final reported risk is at least 10^{-5}. Both choices are about *reported risk*; neither is a collision label.

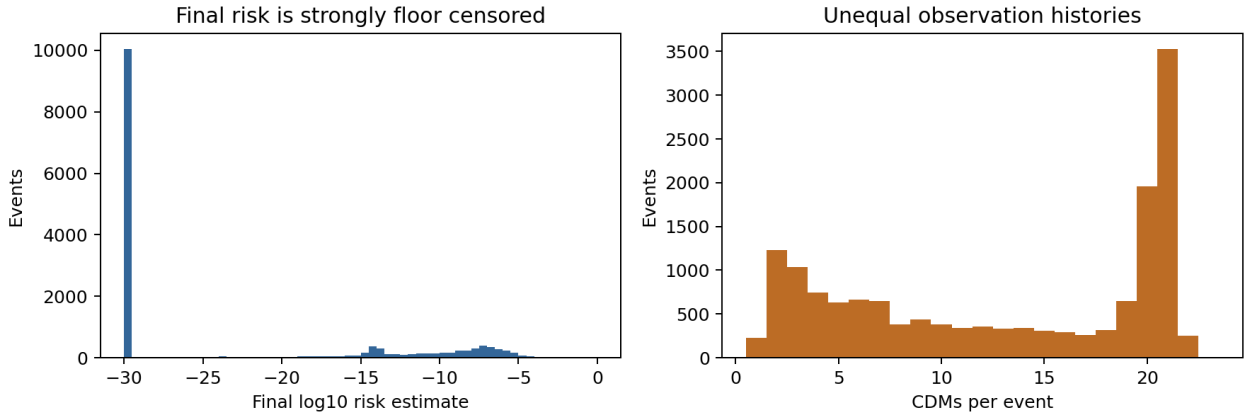
How the comparison was set up

I made snapshots at 120, 96, 72 and 48 hours before TCA. A snapshot takes the latest CDM available *at or before* that lead-time cutoff and uses only its preceding history. The final CDM must arrive later and within 24 hours of TCA. An encounter with no message available at a cutoff is left out of that horizon. There is no interpolation from later messages.

All snapshots from an encounter stay in one partition: 60% training, 15% validation, 10% calibration and 15% test. This keeps later records from an encounter out of training when an earlier record from that encounter is tested. It also lets me compare the four horizons on the same 1,251 held-out encounters. The final risk category was used to balance these partitions, never as a model input. The validation script checks the cutoff and partition rules against the raw rows.

The raw archive has 103 columns. Models use a smaller set of available geometry, risk, uncertainty and observation-history features; missing numerical values are imputed from training data only.

What the encounter histories look like



Left: final reported log-risk has a large pile-up at the -30 floor. Right: encounters have unequal numbers of CDMs.

The floor dominates this dataset: 65.4% of encounters end at -30. Only 131 of all 15,321 encounters finish at or above 10^{-5}. Among the 11,630 whose last CDM is within 24 hours of TCA, that count is 89. A low average regression error can therefore coexist with poor performance on the rare encounters that matter most to a warning system.

The encounter histories also have different lengths. Some have one CDM; others have as many as 23. Position uncertainty usually contracts as TCA nears, but the risk estimate does not simply move in one direction. On encounters with a near-TCA final message, the median change from first to last *capped* log-risk is -2.67 units. This is a useful overall description, not a rule for individual cases. A revised miss distance or covariance can push an encounter the other way.

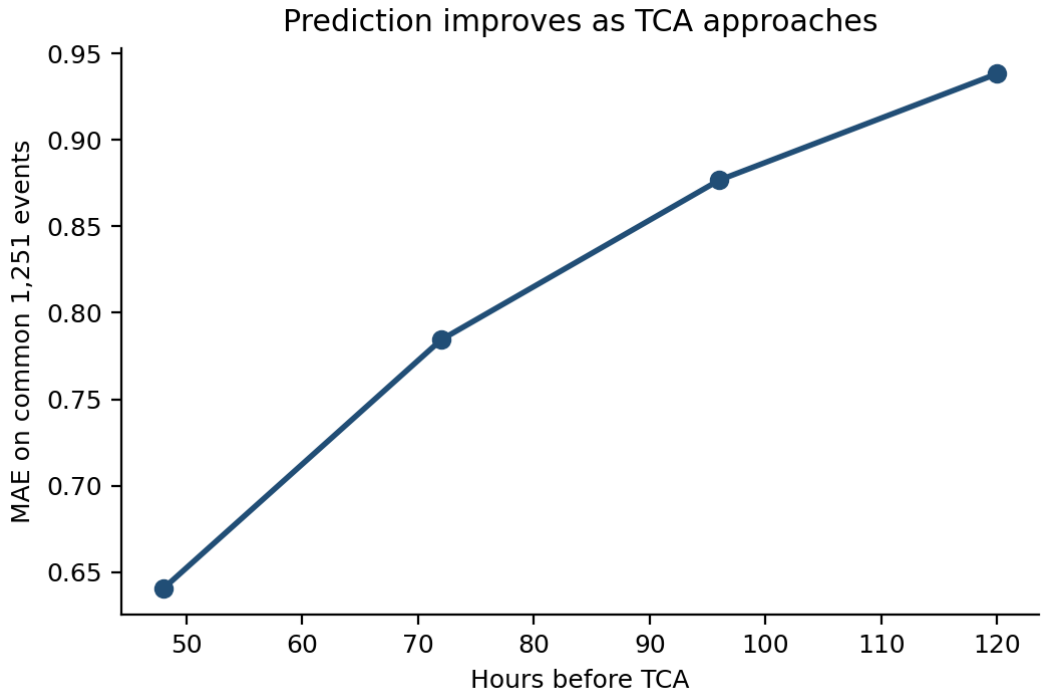
What the statistical models added

I started with linear and logistic models because their coefficients make the assumptions visible. At 72 hours, the linear model explains about 20% of training variation in capped final risk. Current risk and miss distance have measurable associations with final risk, but residual checks show that a straight-line, constant-variance model misses important structure. The tested linear interaction between current risk and uncertainty was not supported by the data. These coefficients describe associations with a later *calculated* estimate; they should not be read as causal effects on collision risk.

Repeated messages from one encounter are related. A random-intercept model on 220 sampled encounters put about 66% of its conditional variation at the encounter level. A smaller Bayesian hierarchy gave a similar qualitative picture for stable, nonfloor histories. This is why treating all 199,082 CDMs as independent rows would be misleading. It also explains the event-level test split.

Missingness deserves attention. The chaser radar cross-section estimate is absent in 32.1% of CDMs, while a cluster of chaser covariance fields is absent in 5.6%. The missingness may reflect what could be observed about an object. A multiple-imputation check did not establish a clear radar-cross-section effect; the main forecasting feature set does not rely on that field.

Forecasting the final estimate



XGBoost on the same 1,251 held-out encounters at each lead time. Lower mean absolute error (MAE) is better.

On those same encounters, XGBoost MAE moves from 0.938 at 120 hours to 0.877 at 96, 0.784 at 72 and 0.640 at 48. Holding the encounters fixed matters: otherwise, later cutoffs include additional cases and an apparent improvement might just reflect a different test population. The result is unsurprising in direction, but it puts a size on the value of more recent tracking data.

At 72 hours, the full eligible test cohort has 1,462 encounters. The compact model comparison below uses that cohort, so its XGBoost value differs from the paired-horizon series above.

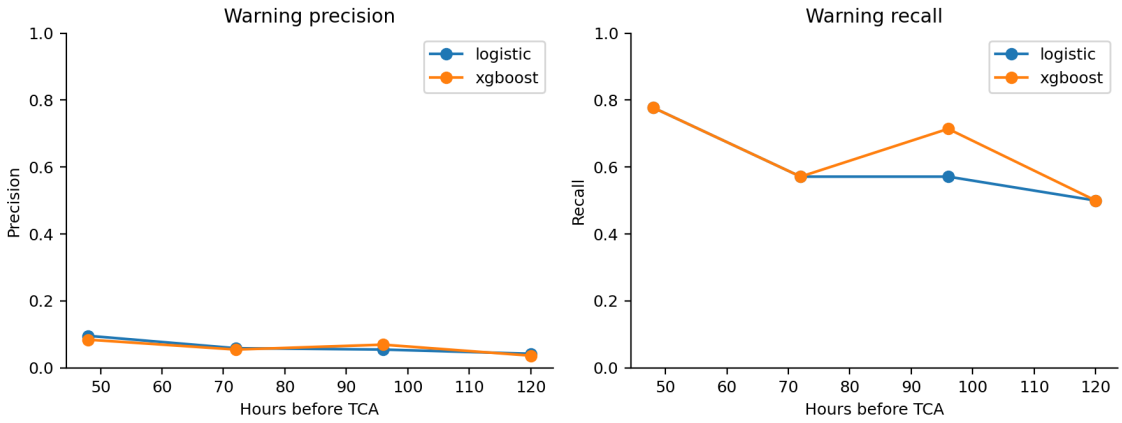
Model	72-hour test MAE
Linear regression	1.022
Random forest	0.828
XGBoost	0.846
GRU sequence model	0.776
LSTM sequence model	0.911

The tree models beat the linear baseline. The XGBoost advantage over linear regression is 0.176 MAE units, with an event-bootstrap 95% interval of 0.144 to 0.205. The GRU did best at 72 hours, though it did not remain best at 48 hours, where the forest reached 0.691 MAE and the GRU 0.739. Model choice therefore depends on the lead time; added sequence complexity did not give a consistent win.

MAE is measured in $\log_{10}$ risk units after the -15 cap. It is useful for comparing point forecasts under the same definition, but it says little by itself about whether the model can issue a practical warning. Most final values are near the floor, so the next page checks the rare high-risk category separately.

Warnings from rare cases

72-hour XGBoost warnings	Held-out count
High-risk encounters found	4
High-risk encounters missed	3
False alarms	69



Warning precision stays low at every cutoff. Recall varies sharply because each test cohort contains only a few high-risk encounters.

The warning rule labels an encounter positive if its final *reported* risk reaches 10^{-5}. At 72 hours, only seven held-out encounters did. The XGBoost rule found four of them, missed three and raised 69 false alarms. Precision was 5.5%: most alerts did not end in the high-risk category. Lowering a threshold to catch more rare positives typically means reviewing more encounters that do not cross it. The counts are more informative here than a headline percentage.

The threshold was selected on the calibration set, not adjusted after seeing the test set. Even so, this is a small-positive-count result. It cannot establish how many alerts an operational system would create. The ESA archive was screened before release, and the separate competition test set was deliberately enriched for high-risk cases. In that official split, the high-risk share is about nine times the eligible training share; precision measured there cannot be carried over to the raw archive without accounting for that shift.

Uncertainty, takeaways and limits

How certain are the forecasts?

I used a separate calibration partition to form 90% split-conformal intervals around the point forecasts. Their held-out marginal coverage was roughly 90–92%. That is useful evidence for the event-level test population, although some intervals are broad: a well-covered interval can still be too wide to guide a decision.

The focused Bayesian regression tells a different story. It models only encounters whose final risk lies above the -15 floor. Its sampler diagnostics were acceptable, and its point MAE was 1.463 on 205 held-out nonfloor encounters, but its nominal 90% posterior predictive intervals covered only 79% of them. Good chain behaviour does not guarantee that the predictive distribution matches the data. This model is also conditional on a selected subset, so its score should not be compared directly with full-cohort MAE.

Those two interval results answer different questions. Conformal coverage is marginal over eligible held-out encounters under an exchangeability assumption. The Bayesian interval describes a conditional continuous-risk model on nonfloor cases. Neither validates uncertainty for a new satellite, a later time period or physical collision occurrence.

What I would take from this project

The clearest result is that recent CDMs improve forecasts of the last risk estimate, and nonlinear models capture patterns that the simple linear baseline misses. The more consequential result is the warning trade-off: a model can have decent average error while generating many false alarms when the high-risk category is extremely rare. Reporting both error and the underlying alert counts changes how the model should be judged.

The statistical work was useful even where it did not produce the winning predictor. The event-level variance showed why repeated CDMs need grouped analysis. Residual checks exposed weaknesses in the linear model. The Bayesian fit showed why convergence is only a starting point for checking uncertainty. Calibrating intervals on untouched encounters brought marginal coverage much closer to its nominal target, though not necessarily to a useful width.

Limits I would keep in view

There are no observed collision labels. The target is a later risk calculation that shares physical inputs with the predictors. The -30 software floor and my -15 regression cap change what MAE means. Dates and spacecraft identities are anonymised, so I could split by encounter but could not test a forward-time deployment or a genuinely unseen satellite. The archive itself is selected; the official competition test distribution is selected differently again. The high-risk test counts are small enough that warning performance could shift sharply with a few new cases.

A stronger next step would be a model that handles the floor explicitly, followed by validation on later encounters with spacecraft identities and a representative warning workload. Until then, these results support comparison of *reported risk forecasts*, not an operational collision-avoidance claim.

Files and reproduction

The repository has the numbered scripts, saved figures, result tables, model states and this document's LaTeX source. The raw ESA files are downloaded separately. `README.md` gives the exact directory layout and run order; `scripts/08_validate.py` checks cutoff and split invariants. The main source is the ESA dataset; the column descriptions and challenge overview explain its selection and original task.